Auteur: Anass Hamzaoui
Studierichting: Informatica
Oefeningenreeks 2D nr. 16.10

$$
\begin{aligned}
&\lim_{x\to\frac{\pi}{2}} \frac{1-\cos 4x}{(2x-\pi)\cdot\sin 2x} \\
&= \lim_{x\to\frac{\pi}{2}} \frac{1-1+2\cdot\sin^2 2x}{(2x-\pi)\cdot\sin 2x} \\
&= \lim_{x\to\frac{\pi}{2}} \frac{2\cdot\sin^2 2x}{(2x-\pi)\cdot\sin 2x} \\
&= \lim_{x\to\frac{\pi}{2}} \frac{2\cdot\sin 2x}{(2x-\pi)} \\
&= \lim_{x\to\frac{\pi}{2}} \frac{-2\cdot\sin(2x-\pi)}{(2x-\pi)} \\
&= -2
\end{aligned}
$$

References